Imagine, for instance, that you are busily calculating the physical properties—particle masses and force charges—associated with one possible Calabi-Yau choice for the extra dimensions. You are not particularly concerned with matching your detailed results with experiment, since as we have seen a number of theoretical and technological obstacles make doing this quite difficult at present. Instead, you are working through a thought experiment concerned with what the world would look like if a particular Calabi-Yau space were selected. For a while, everything is going along fine, but then, in the midst of your work, you come upon a mathematical calculation of insurmountable difficulty. No one, not even the world's most expert mathematicians, can figure out how to proceed. You are stuck. But then you realize that this Calabi-Yau has a mirror partner. Since the resulting string physics associated with each member of a mirror pair is identical, you recognize that you are free to do your calculations making use of either. And so, you rephrase the difficult calculation on the original Calabi-Yau space in terms of a calculation on its mirror, assured that the result of the calculation—the physics—will be the same. At first sight you might think that the rephrased version of the calculation will be as difficult as the original. But here you come upon a pleasant and powerful surprise: You discover that although the result will be the same, the detailed form of the calculation is very different, and in some cases the horribly difficult calculation you started with turns into an extremely easy calculation on the mirror Calabi-Yau space. There is no simple explanation for why this happens, but—at least for certain calculations—it most definitely does, and the decrease in level of difficulty can be dramatic. The implication, of course, is clear: You are no longer stuck.

It's somewhat as if someone requires you to count exactly the number of oranges that are haphazardly jumbled together in an enormous bin, some 50 feet on each side and 10 feet deep. You start to count them one by one, but soon realize that the task is just too laborious. Luckily, though, a friend comes along who was present when the oranges were delivered. He tells you that they arrived neatly packed in smaller boxes (one of which he just happens to be holding) that when stacked were 20 boxes long, by 20 boxes deep, by 20 boxes high. You quickly calculate that they arrived in 8,000 boxes, and that